

*Differentiable Cinematic
Rendering: Optimizing Lens
and Lighting Parameters for
Stylized Image Synthesis*

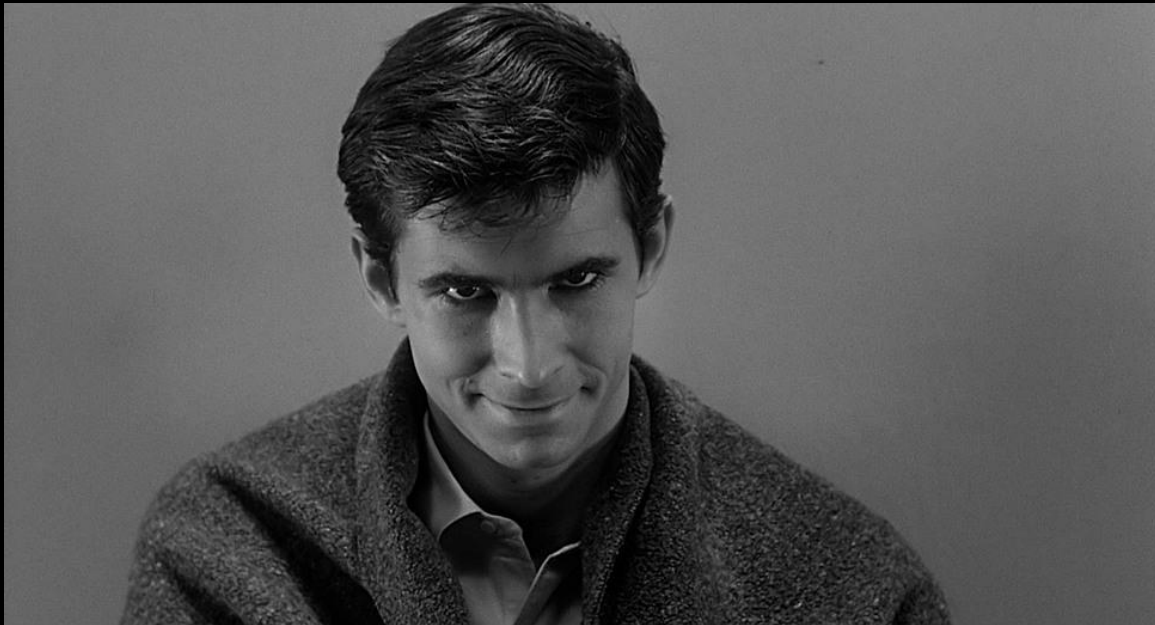


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CSPB 4502 - Data Mining

Project Questions

- ❑ What optical characteristics define Psycho vs. Raging Bull?
 - *Can differentiable rendering recover film-specific lens behavior?*
 - *Can perceptual losses reveal stylistic 'optical DNA'?*



Data Preparation

- ❑ Collected reference frames from *Psycho* & *Raging Bull*.
 - *Converted frames to 128×128 grayscale.*
 - *Normalized luminance, fixed seeds for reproducibility.*
 - *Prepared minimal geometry scenes for inverse rendering.*



Tools Used

- ❑ Python 3.10 – core scripting
 - *PyTorch + PyTorch3D – differentiable rendering & optimization*
 - *LPIPS model – perceptual loss computation*
 - *Matplotlib – analysis & visualization*
 - *VS Code + GitHub – workflow management*



Techniques Applied

- ❑ Differentiable rendering pipeline
 - *Optimization of lens distortion, lighting, camera pose*
 - *Loss: $\alpha \cdot \text{MSE} + \beta \cdot \text{LPIPS} + \gamma \cdot (1 - \text{SSIM})$*
 - *Gradient-based parameter recovery (Adam)*
 - *Comparative analysis across film-historical targets*



Knowledge Gained

- ❑ Recovered subtle, film-specific optical tendencies
 - *Demonstrated that minimal scenes can yield meaningful lens signatures*
 - *Showed perceptual optimization as a tool for stylistic analysis*
 - *Applications: virtual cinematography, restoration, style transfer*



Core Findings

❑ Minimal Rendering, Meaningful Style

- A deliberately minimal differentiable renderer—reduced to only light, lens, and a single surface—was still able to recover **film-specific optical signatures**.
- Even with 128×128 grayscale inputs and a simple sphere geometry, the system converged toward lens behaviors historically associated with *Psycho* and *Raging Bull*.
- This demonstrates that a film’s visual “feel” is deeply encoded in its optical pathway, independent of scene complexity.
- Distortion tendencies, tonal rolloff, and optical fingerprints that provide quantifiable traces detectable through gradient-based optimization.
- Even at minimal dimensional scale, first-order parameters reveal meaningful stylistic identity.



Conclusion

- ❑ This work serves as a proof-of concept for analyzing film style through optimization rather than imitation.
- ❑ This approach bridges computational reasoning with film-historical intuition that opens possibilities for:
 - *Data-driven lens emulation*
 - *Virtual cinematography informed by historical optics*
 - *Pedagogy grounded in measurable properties*



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